

COMMENTARY

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Highlights of recent cancer research

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1 Introduction

Over the past few months, there has been significant progress in the cancer research field. This article intends to present a brief review of representative breakthroughs.

1.1 Cancer hallmarks and statistics

In the 2022 January Issue of *Cancer Discovery*, Douglas Hanahan published the third version of “Hallmarks of Cancer: New Dimensions” [1]. As he indicated, “The Hallmarks of Cancer were proposed as a set of functional capabilities acquired by human cells as they make their way from normalcy to neoplastic growth states, more specifically capabilities that are crucial for their ability to form malignant tumors”. In 2000 and 2011, Robert Weinberg and Douglas Hanahan published the first and second versions of the review article [2, 3]. In the first version of the review article [2], the authors proposed six hallmarks of cancer: “self-sufficiency in growth signals”, “insensitivity to anti-growth signals”, “apoptotic evasion”, “limitless replicative potential”, “sustained angiogenesis” and “tissue invasion and metastasis” [2]. Four extra characteristics were added in the second version of the article. Two features, “deregulating cellular energetics” and “avoiding immune destruction”, were included as emerging hallmarks. Additionally, “tumor-promoting inflammation” and “genome instability and mutation” were included as enabling hallmarks [3]. In the most recent version of “Hallmarks of Cancer”, Douglas Hanahan added another four hallmarks and enabling characteristics: “unlocking phenotypic plasticity”, “nonmutational epigenetic reprogramming”, “polymorphic microbiomes”, and “senescent cells” [1]. Each successive inclusion into the expanding body of work is helping to distill the complexity of cancer into an increasing logical science and facilitating a deeper

understanding of the physiological mechanisms associated with cancer, as well as identifying features that can be integrated into clinical practice.

Another important event in the cancer research field is the release of a report from the American Cancer Society entitled: “Cancer Statistics 2022” [4]. In 2022, 1,918,030 new cases of cancer are projected to occur in the USA. Between 1991 and 2019, cancer-related deaths dropped by 32% in the USA. This success is largely due to cancer prevention measures; for instance, the reduction in tobacco smoking has resulted in declining rates of lung cancer and other smoking-related cancers. Other contributing factors are advances in early detection, targeted therapy, surgical therapies. The adjuvant chemotherapies for colon and breast cancer, and advanced therapies for many cancers, have also contributed to this reduction [4].

1.2 Structure biology

Last December, two articles describing the structure of Anaplastic Lymphoma Kinase (ALK) were published in *Nature* [5, 6]. ALK and the related Leukocyte Tyrosine Kinase (LTK) play critical functional roles in neural development, cancer, and autoimmune diseases. ALK fusion proteins and mutated proteins have been identified as targets of therapeutic relevance for cancer and other diseases. The US Food and Drug Administration (FDA) has approved several ALK Inhibitors as drugs for lung cancer, including Crizotinib, Ceritinib, Alectinib, and Brigatinib. Steven De Munck et al. [5] reported that the cytokine-binding segments of human ALK and LTK comprise a novel architectural chimera of a permuted TNF-like module. The binding of the cognate cytokines, ALKAL1 and ALKAL2, to ALK and LTK elicit similar dimeric assemblies with two-fold symmetry. The extracellular domain of ALK is rather large, including two NAN domains, one LDL α domain, and a glycine-rich protein. The glycine rich domain (GRD) is sufficient for ligand binding to regulate receptor activity. Tongqing Li et al. [6] reported the crystal structure of the ALK GRD.

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The authors showed that repetitive glycine residues in the GRD form rigid helices that separate the major ligand-binding site from a distal polyglycine extension loop (PXL). The PXL of one receptor functions as a sensor for the complex by interacting with a ligand-bound second receptor. These two articles provide high-resolution structures of the extracellular domain of ALK, explain how ALK activity is modulated, and suggest new therapeutic opportunities for ALK-related cancers [5, 6].

1.3 Cancer biology

Somatic mutations within driver genes in the human genome are believed to be instrumental for cancer initiation and development. Interestingly, B Colon et al. reported that the majority of newly formed esophageal cancers are eliminated through competition with mutant clones in the adjacent normal tissue. The group found that most pre-neoplastic lesions were eliminated with no indication of cell death, anti-cancer immune response, or decreased cell proliferation. Mutant clones in normal epithelium have anti-cancer activity and eliminate emerging cancer formation [7].

In last October's issue of *Cell*, Qingxu Liu et al. reported a novel mechanism of liver cancer development; glycogen accumulation and phase separation drive liver cancer initiation. They found that cancer-initiating liver cells adopt a glycogen storage state in order to improve survival. They reported that the downregulation of glucose-6-phosphatase (G6PC), which catalyzes the final step of glycogenolysis, augments glycogen storage and neoplastic cell transformation. Glycogen accumulation undergoes liquid-liquid phase separation, resulting in the inhibition of HIPPO signaling. More importantly, elimination of glycogen accumulation abrogates liver carcinogenesis and cancer incidence [8].

The mTOR complex 1 (mTORC1) controls cell metabolism and cell growth by sensing amino acid levels. Jie Chen et al. reported a leucine sensor to regulate mTORC1 signal transduction pathway [9]. The authors identified that SAR1B, a small GTPase, controls leucine-dependent GATOR's regulation of mTORC1 signaling. SAR1B can directly bind with leucine via V100, A141 and E144. More importantly, SAR1B and sestrin2 are two different sensors of leucine. They recognized different structural features of leucine and binds various subunits of GATOR2. In an animal model, the authors found that SAR1B is a tumor suppressor for lung cancer. The knockdown of SAR1A and SAR1B increases the mTORC1-dependent cancer growth in mice [9].

1.4 Microbiota and Cancer

Gut microbiota affects cancer cell growth, anti-cancer immunity, and cancer cell response to chemo- or immunotherapy. Recently, Khiem C. Lam et al. [10] and

Christine N. Spencer [11] independently reported that a high-fiber diet affects microbiota and triggers immunity for anti-cancer. Spencer's report indicated that the higher fiber diet was associated with improved progression-free survival in 128 patients on immune check-point blockade (ICB) [11]. The best benefit was observed in patients with a high fiber diet and no probiotic use. In preclinical mouse models, they found an impaired treatment response to anti-programmed cell death therapy in mice receiving a low fiber diet or probiotics [11]. Lam et al. showed that consumption of high fiber or transferring the fecal microbiota from ICB responder melanoma patients can trigger STING-type 1 IFN-dependent monocyte reprogramming pathway. The group highlighted a mechanistic link between the innate tumor microenvironment and microbiota that could be harnessed for more efficacious cancer therapies [10].

1.5 Clinical trial: KEYNOTE-811

Yelena Y. Janigian et al. reported the result of the randomized, double-blind, placebo-controlled phase III KEYNOTE-811 study of pembrolizumab plus trastuzumab and chemotherapy conducted with 434 HER-2 positive gastric or gastro-esophageal junction adenocarcinoma patients. They reported that combining pembrolizumab plus trastuzumab and chemotherapy could decrease tumor size and improve objective response; induction of a complete response was observed in several patients [12].

1.6 Multi-omic in Cancer research

Neoadjuvant therapy prior to surgery is increasingly used in many types of human cancers; however, only a subset of patients respond favorably. Stephen-John Sammut et al. used tumor tissues derived from patients prior to treatment to generate multi-omic data that was utilized for subsequent machine learning application [13]. The group collected breast cancer biopsies from 168 patients treated with chemotherapy+/- HER2-target therapy prior to surgery. Multi-platform profiling (DNA, RNA sequencing and digital pathology) of these samples was subsequently analyzed. They found that response to treatment is modulated by the pre-treated tumor parameters and that multi-omics can be integrated into predictive models using machine learning. This system can be used as a predictor of breast cancer therapy response. Such an approach may be used for predictors of other cancers [13].

Metastatic cancers are the main cause of cancer patients' death. By analyzing genomic and clinic data from 25,000 patients, Bastien Nguyen et al. discovered chromosomal instability is correlated with metastatic burden in many cancers including lung adenocarcinoma, prostate adenocarcinoma, HR⁺/HER2⁺ breast ductal

carcinoma, but not in colon cancer, high-grade ovarian cancer [14]. Oncogene mutations and alterations are increased in metastases of many cancers. The majority samples were obtained from metastatic lymph nodes ($n = 2305, 23\%$), liver ($n = 2289, 23\%$), lung ($n = 982, 10\%$), or bone ($n = 726, 7\%$). The most common target organs for metastasis of 50 types of cancer were liver, lung and bone. Higher chromosomal instability was associated with increased metastatic burden. Very interestingly, they found specific recurrent oncogenic alterations associated with metastasis patterns. For example, prostate cancer bone metastasis had higher frequency of AR amplification, PTEN deletion and lower frequency of ERG fusions, when prostate cancer metastasis in liver had a higher PTEN loss, RB1 loss and APC mutations; prostate cancer with brain metastasis had high frequency of AR amplification, NOTCH alterations; and when prostate cancer with lung metastasis had a higher frequency of APC mutation, CTNNB1 mutation, WNT signaling activation. HR⁺/HER2⁻ breast cancer with liver metastasis had higher ESR1 mutations; breast cancer with bone metastasis had a lower frequency of CEBF mutations, while PI3K changes were higher in those breast cancer bone metastases. HR⁺/HER2⁻ breast cancer patients with brain metastasis had a lower MAP3K1 mutation rate, lower TP53 rate and higher PTEN mutations. Esophageal cancer with lung metastasis had higher frequency of ERBB2 amplification [14].

1.7 Cancer immunology

Cancer immunotherapies targeting tumor-specific T cells works very well in certain patients, but not in other patients. The major reason for such insensitivity is T cell exhaustion, a dysfunctional state of T cells. In the tumor microenvironment (TME), tumor-infiltrating lymphocytes play crucial roles. In order to compare the heterogeneity and dynamics of tumor-infiltrating T lymphocytes among cancers, Liangtao Zheng et al. established a single-cell RNA-sequencing pan-cancer atlas of T lymphocytes of 316 patients. They studied multiple potential tumor-reactive T cell population among 21 types of cancer. After comparing the frequencies of CD8⁺ terminal exhausting T cells in all types of cancer, the authors discovered multiple state-transition paths and the preference of those paths among different tumor types. The immune-typing based on T cell composition within tumors could be used for classifying cancers into different groups with clinical trait specificity. Such results may facilitate the development of effective therapy and diagnosis method for cancer [15].

Acknowledgments

Not applicable.

Author's contributions

The paper was authored by Prof. Zigang Dong. The author(s) read and approved the final manuscript.

Funding

There is no funding source to declare.

Availability of data and materials

Not applicable.

Declarations

Ethics approval and consent to participate

I have read and understood your journal's policy, and I believe that neither the manuscript nor the study violates any of these.

All author consent to the participation.

Consent for publication

All author consent to the participation.

Competing interests

All authors declare that they have no conflict of interest.

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Received: 22 February 2022 Accepted: 15 March 2022

Published online: 28 April 2022

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